Journal of Imaging Informatics in Medicine (2025) 38:775–792
https://doi.org/10.1007/s10278-024-01209-4

ORIGINAL PAPER

Deep Learning-Based Model for Non-invasive Hemoglobin Estimation via Body Parts Images: A Retrospective Analysis and a Prospective Emergency Department Study

En-Ting Lin[1] · Shao-Chi Lu[1] · An-Sheng Liu[1] · Chia-Hsin Ko[2] · Chien-Hua Huang[2] · Chu-Lin Tsai[2] · Li-Chen Fu[1]

Received: 6 June 2024 / Revised: 15 July 2024 / Accepted: 22 July 2024 / Published online: 19 August 2024
© The Author(s) under exclusive licence to Society for Imaging Informatics in Medicine 2024

Abstract
Anemia is a significant global health issue, affecting over a billion people worldwide, according to the World Health Organization. Generally, the gold standard for diagnosing anemia relies on laboratory measurements of hemoglobin. To meet the need in clinical practice, physicians often rely on visual examination of specific areas, such as conjunctiva, to assess pallor. However, this method is subjective and relies on the physician's experience. Therefore, we proposed a deep learning prediction model based on three input images from different body parts, namely, conjunctiva, palm, and fingernail. By incorporating additional body part labels and employing a fusion attention mechanism, the model learns and enhances the salient features of each body part during training, enabling it to produce reliable results. Additionally, we employ a dual loss function that allows the regression model to benefit from well-established classification methods, thereby achieving stable handling of minority samples. We used a retrospective data set (EYES-DEFY-ANEMIA) to develop this model called Body-Part-Anemia Network (BPANet). The BPANet showed excellent performance in detecting anemia, with accuracy of 0.849 and an F1-score of 0.828. Our multi-body-part model has been validated on a prospectively collected data set of 101 patients in National Taiwan University Hospital. The prediction accuracy as well as F1-score can achieve as high as 0.716 and 0.788, respectively. To sum up, we have developed and validated a novel non-invasive hemoglobin prediction model based on image input from multiple body parts, with the potential of real-time use at home and in clinical settings.

Keywords Hemoglobin Estimation · Anemia Detection · Deep Learning · Computer Vision

Introduction

Anemia, a significant global health issue, involves a reduction in red blood cell count or hemoglobin concentration in the blood. Symptoms encompass dizziness, paleness, breathlessness, and palpitations. Severe cases may lead to excessive sleepiness, reduced appetite, cardiomegaly, and even heart failure. Anemia symptoms often signal underlying diseases, emphasizing the need for early detection to improve patient health.

The gold standard for diagnosing anemia is the laboratory measurement of hemoglobin concentration in the blood, which requires specialized equipment for blood sampling. However, this process can lead to delays in initiating patient treatment. To quickly screen for anemia, clinicians often rely on physical examinations. One approach involves identifying symptoms such as skin pallor in areas less influenced by pigmentation and where blood vessels are more visible, such as the conjunctiva, fingernails, and palms.

Although some studies have demonstrated the effectiveness of physical examinations [1], it is subjective and relies heavily on physician experience. Factors such as patient age, comorbid conditions, skin color, and lighting conditions at the examination site can also affect the assessment of rosy coloration, making it difficult for

Chu-Lin Tsai and Li-Chen Fu contributed equally.

✉ Chu-Lin Tsai
chulintsai@ntu.edu.tw

✉ Li-Chen Fu
lichen@ntu.edu.tw

[1] Department of Computer Science and Information Engineering, National Taiwan University, CSIE Der Tian Hall No. 1, Sec. 4, Roosevelt Road, Taipei 106319, Taiwan

[2] Department of Emergency Medicine, National Taiwan University Hospital and National Taiwan University College of Medicine, 7 Zhongshan S. Rd, Taipei 100, Taiwan

non-experts to practice. In recent years, with advances in hardware and the remarkable performance of machine and deep learning in computer vision, researchers have attempted to improve physician judgments using these methods. These approaches typically input images of common body parts to determine if the patient is anemic or even predict their hemoglobin concentration.

Since the advent of convolutional neural networks (CNNs), deep learning techniques have been prevalent in resolving computer vision tasks, including several medical applications. Unlike early machine learning methods, researchers no longer need manually designed features. Instead, they provide appropriate cues to the model through various modules and loss functions, allowing it to learn the most discriminative features from the raw image data. However, few studies have applied these newer techniques to anemia detection, suggesting that there is still room for further research and improvement in this task.

Existing studies have specialized models for single body part images. This means that once a model is trained on a dataset of a particular body part, it cannot predict results for other body parts. However, clinicians sometimes may rely on using different body parts which are available for anemia assessment, and the manifestations could differ between different body parts. In addition, some patients may not be able to provide specific body part images for reasons such as the diseases themselves. Therefore, for practical applications, it is necessary to develop an approach that can incorporate images from different body parts.

To overcome the aforementioned issues, we proposed a deep-learning-based model called Body-Part-Anemia Network (BPANet). The model takes the image of the body part as its primary input, complemented by equally important features such as gender and age. Furthermore, to solve the problem of predicting only a single body part, the model incorporates a body part tag as an additional input.

Related Work

Non-invasive Anemia Detection (Mathematical Models)

Due to the time-consuming and costly nature of traditional invasive detection methods, some studies have turned to develop non-invasive prediction methods using images or videos of the target body parts. Suner et al. [2] separated the RGB channels of the input images and applied an algorithm to compute and output predictions of hemoglobin concentration. Although their results demonstrated the effectiveness of computer vision algorithms in anemia detection, the method relied heavily on the accuracy of RGB separation, which might be affected by varying lighting conditions. Tamir et al. [3] determined whether a user had anemia by performing thresholding on the red and green components of the images, but this approach was simplistic and might not capture the complexity of image input variations. Khan et al. [4] pioneered a hardware-level prediction approach by using an FPGA-based architecture combined with a custom linear regression model for blood flow-based prediction. While innovative, the hardware dependency limits its accessibility and scalability. Collings et al. [5] calculated the erythema index (EI) from conjunctival images and analyzed its correlation with hemoglobin concentration. However, the reliance on a single feature may limit the prediction accuracy. Chen et al. [6] further extended the use of EI by incorporating modified Kalman Filtering (KF) and achieved good predictive results on high-risk and low-risk samples through penalty regression. This approach, however, was still limited by the chosen feature set. Anggraeni et al. [7] used self-collected conjunctival images of pregnant women and established a linear regression model after preprocessing for brightness adjustment. The model's performance was somewhat constrained by the quality of preprocessing steps. These studies introduced various computer vision techniques to the task of anemia detection through simple mathematical models. Many of these methods were dependent on specific preprocessing steps and limited feature sets, which may not handle varying image qualities well. To address these limitations, we sought to integrate feature extraction directly into the learning pipeline, reducing dependency on manual preprocessing and enhancing the generalizability and accuracy of predictions.

Machine-Learning-Based Methods

With advances in machine learning methods such as support vector machines (SVM) and random forest (RF), researchers have replaced manually defined mathematical models with these new methods. These authors defined and computed image features based on their background knowledge and trained these machine-learning models using various datasets. Dimauro et al. [8] used conjunctival images as model input and employed RUSBoost to handle unbalanced data. However, the feature extraction in this study was manual, potentially missing critical information. Appiahene et al. [9] first transformed the input palm images into the CIELab color space and then utilized kNN, SVM, and Naïve Bayes for prediction. This transformation step, while helpful, added complexity and potential sources of error. Asare et al. [10] further compared images from various body parts and concluded that palm images were best for prediction. However, this study was a comparative study without integrating information from multiple body parts. Wang et al. [11] developed HemaApp based on support vector regression (SVR), allowing users to self-test hemoglobin levels using smartphone

cameras. However, the system's effectiveness was limited by its dependency on a specific device and light source, which affected its generalizability. Meanwhile, the application developed by Mannino et al. [12] took nail images as input and applied data preprocessing techniques to remove glare noise, thereby improving prediction accuracy. However, the application's performance might be affected by conditions causing nailbed discolorations. The aforementioned methods often involved complex preprocessing steps and relied on specific features. They also handled feature extraction and prediction separately, which may lead to integration issues. We sought to simplify the workflow by processing images in their original color/space and integrating feature extraction and prediction within a single, unified framework.

Deep-Learning-Based Methods

In recent years, various deep learning models have been widely applied to various computer vision tasks, with CNN-based models being predominant, especially in their earlier developments. The advantage of this approach is that the model can learn how to extract the most representative features through the process of backpropagation. Zhang et al. [13] utilized face videos as input and extracted facial images through a face detection model. They employed CNN models such as MobileNet and ResNet for anemia classification and achieved higher accuracy compared to doctor's prediction. However, their study focused on face videos, which might not generalize well to other body parts. Zheng et al. [14] proposed the GLUDA model, which is based on EfficientNet. The model took both global eye images and the region of interest (ROI) of the conjunctiva as inputs and achieves promising results. While promising, the algorithm had a higher requirement for image quality. Roychowdhury et al. [15] deployed a trained CNN model on the Azure cloud platform and developed a web-based GUI service. However, cloud dependency can be a barrier for some users. Kesarwani et al. [16] performed feature extraction on palm videos using different regression models. They then fed these features into a multi-layer perceptron to predict hemoglobin concentration. The separation of feature extraction and prediction may introduce integration issues. These deep learning models allow researchers to reduce excessive design in the method and achieve improved accuracy compared to the previously mentioned methods. The aforementioned examples often focused on specific body parts, limiting the generalizability. We designed our model to include images of multiple body parts, integrate demographic information, and employ a dual loss function with the goal to increase generalizability and enhance accuracy.

Methods (BPANET)

System Overview

The overview of the designed model is shown in Fig. 1. We primarily used images of body parts as input for the non-invasive prediction of hemoglobin concentration. In addition to the image input, gender and age were included as part of the input, considering their importance for the tasks [17]. Since these demographic variables were readily available, the advantages of including them outweigh the disadvantages. Furthermore, it was worthwhile to emphasize that one of the main contributions

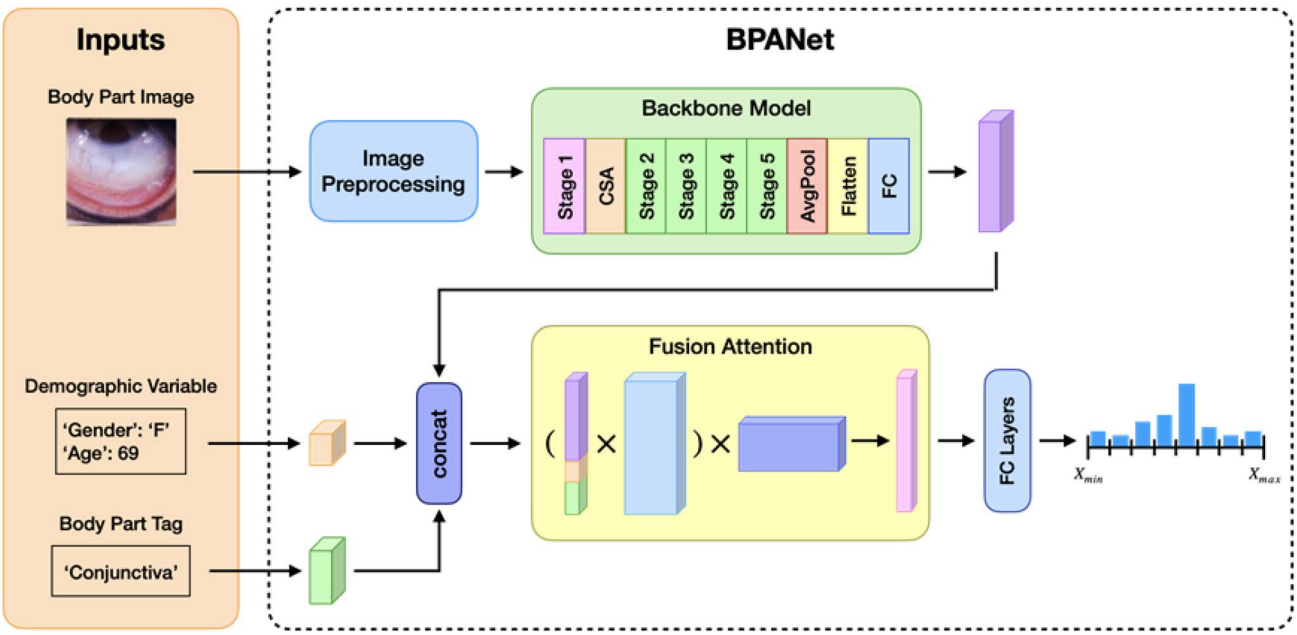


Fig. 1 The overview of the proposed BPANet model. CSA indicates the Channel-Spatial Attention Module

of this research work was that the tag of the body part was also included as input to the model. The details of this aspect are described further in the following section.

Due to the presence of unwanted reflective interference in certain body parts, such as the conjunctiva and nails, the image data needed to undergo a preprocessing step to remove such noise. In order to extract the effective features from the rectified image data, our proposed model decided to use ResNet50 as a backbone model for feature extraction. Besides that, we inserted the Channel-Spatial Attention (CSA) module between stage 1 and stage 2 to improve the model's interpretability and feature extraction capability. It is worth mentioning that due to the body-part-aware design in our model, the features extracted by the backbone model represent the common features among all body parts. These features were then fused with features arising from demographic variables and body part tags. Through the fusion attention mechanism, the model refined the key image features specific to each corresponding body part. Finally, the refined features were passed through fully connected layers to produce the model's prediction results.

Preprocessing

For our hemoglobin concentration prediction task, reflections that appeared in the images can introduce significant noise that adversely affects the performance of the model. For example, reflections caused by secretions on the conjunctiva or light shining on the nails often obscured important information in the images, leading to degraded model performance. The two types of reflections mentioned above manifested themselves primarily as specular highlights in the form of bright spots. To remove this noise, we performed the morphological operation known as opening in image processing. The opening operation consists of erosion followed by dilation, which can remove small structures and noise from the image while preserving larger connected regions. Specifically, we used a 3×3 square as the structuring element and performed two iterations to obtain the acceptable processed image. Note that although the processed image was slightly blurred, this did not affect the performance of the model since the image would be downsampled to a lower resolution before it is input to the model.

Channel-Spatial Attention Module

Most current research based on conjunctival images uses segmented, crescent-shaped regions of interest (ROI) as model inputs. However, this approach has several drawbacks. First, it is not conducive to dataset construction, as it requires expert annotators from the relevant domain to label the corresponding ground truth. Even though there are studies that have developed segmentation models specifically for the conjunctival region [18], the prediction errors of these models can introduce undesirable noises into the dataset. Second, considering that our model aims to enable users with no domain knowledge to perform self-detection, the simplest and most effective approach in practical applications should be to allow users to capture images of their own eyes.

To improve the usability of the model while maintaining its accuracy, it is necessary to strengthen the model's ability to extract ROI features. We incorporated a spatial attention mechanism to compute the importance weights for each spatial position, thereby "guiding" the model to focus on relevant regions. On the other hand, in a convolutional neural network, each channel can be considered to represent a specific feature. The channel attention mechanism can adaptively adjust the importance of each channel by computing importance weights for each channel. Therefore, we introduced a channel-spatial attention mechanism into the backbone model to improve the feature extraction capability of the model.

Following the design of CBAM [19], we sequentially applied the channel and spatial attention mechanisms to the input feature map, as shown in Fig. 2. We inserted this module between stage 1 and stage 2 of the backbone model to improve the extraction of low-dimensional features. Regarding the insertion position, we had two considerations. First, due to the limited data, we could not increase the number of training parameters too much. According to experimental results, adding a second attention module leads to overfitting and decreases the model performance. Second, in our opinion, the modeling should be focused on fine-grained information like blood vessels. Therefore, we finally introduced this module at the early stage of the backbone model to meet our design requirements.

Body Part Tag

As mentioned in the previous section, our model design takes into account the real-world application scenarios. While we believe that it might be feasible to simultaneously input the three types of patient body part images into independent modules and then fuse them to generate a prediction result, there is currently no such corresponding dataset available. Even if one would attempt to collect the multi-modal data, the need to train a stable model would result in a significant increase in data requirements. In addition, this would not be a user-friendly design. Imagine if the user has to capture images of the three body parts each time before self-monitoring the anemia situation, then each of these images is essential. This would greatly reduce the user's willingness to use

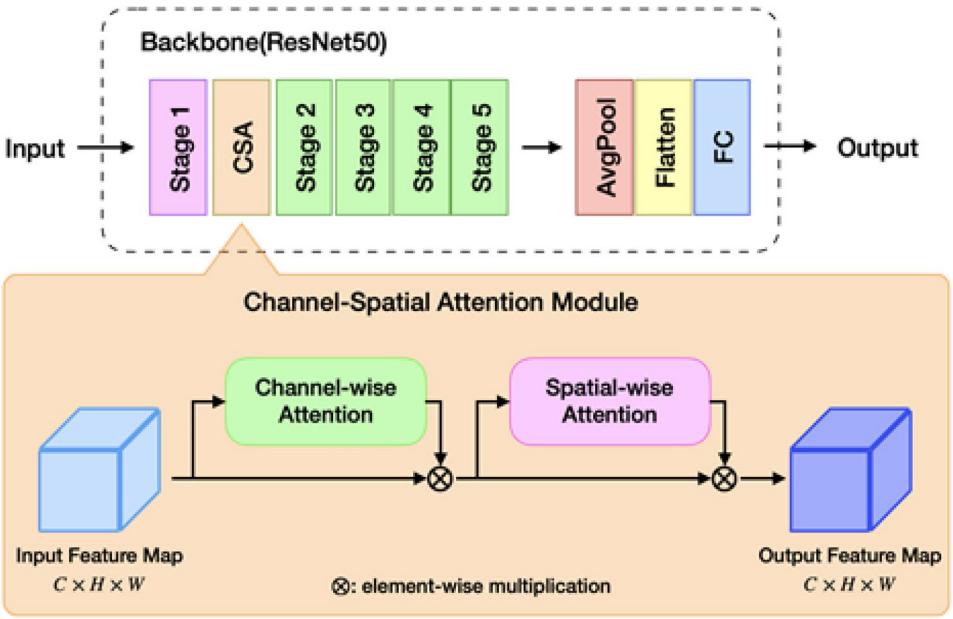


Fig. 2 Channel-Spatial attention module

the system. Therefore, our designed body-part-aware prediction approach utilizes a single model to perform predictions for different body parts. In terms of usage, the user only needs to select the most readily available image of a specific body part and provide the corresponding information to the model for prediction. From the model's perspective, this is equivalent to adding an additive data type to the input, which we call a "body part tag", represented as a one-hot vector of length 4.

Fusion Attention Module

Since gender and age have consistently been significant factors influencing hemoglobin prediction [17], we incorporated these two demographic variables with the image features extracted by the backbone model as part of the features. Furthermore, as mentioned in the previous section, our model is capable of accepting input images of different body parts, allowing for a more flexible prediction approach. To generate appropriate features for different body parts, we enhanced the model's design by introducing an additional body part tag input and designing the Fusion Attention Module, inspired by the application of the self-attention mechanism in feature fusion.

The architecture of the Fusion Attention Module is shown in Fig. 3. Due to the limited parameters of our model caused by the size of the dataset, we were unable to perform computations of quadratic complexity as seen in self-attention

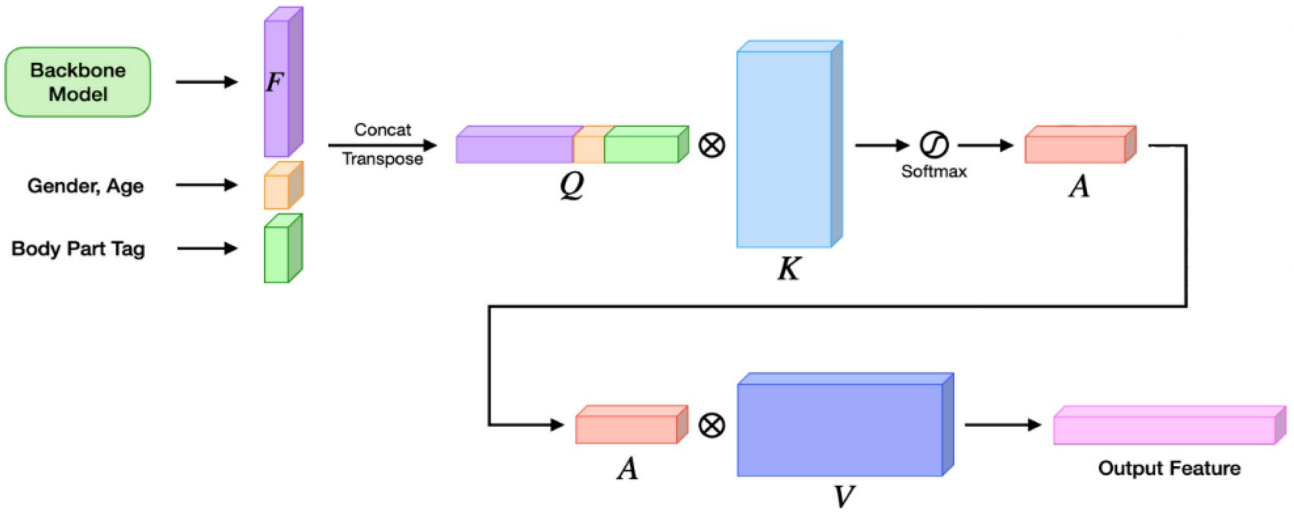


Fig. 3 Fusion Attention Module

mechanisms. In the design of this module, we took inspiration from the idea of the external attention mechanism [20] and modified the computation of the similarity matrix to reduce the computational complexity. In our module, the query matrix Q was equal to the input feature vector F itself, with a shape of $1 \times |F|$, where $|F|$ denoted the length or dimensionality of the feature vector F. The key matrix K differed from the original self-attention mechanism as we made it a learnable matrix with a shape of $|F| \times D$, where D was a small constant. By multiplying Q and K and applying the sigmoid function, we generated a 1D attention matrix A with a length of D. On the other hand, the value matrix V was designed as a learnable $D \times |F|$ matrix. Finally, the output of the module $F\prime$ was obtained by computing $F\prime = A \times Q$.

Dual Loss (DL)

The task of hemoglobin prediction often involves a slight imbalance in data, both in real-world scenarios and in the datasets. At the same time, the concentration of hemoglobin (Hgb) in the human body should not be unbounded. Taking the data we have encountered as an example, the upper and lower bounds of Hgb are 17.8 and 6.7 g/dL, respectively. Thus, considering this property in the model's predictions can significantly reduce the model's convergence time.

Therefore, we employed a Dual Loss as the loss function for training. Before employing the dual loss function, we needed to modify the output of the model. Instead of directly predicting a single value as in previous approaches, the model needs to make predictions in an indirect manner. The procedure is as follows:

1. For any regression task with a finite range of output values$[X_{min}, X_{max}]$, we first select a hyperparameter $N \in N$. Then, we partition the output range into N equally sized bins, such that each bin has a size of $(X_{max} - X_{min}) / N$.
2. The model output is transformed from predicting a single value to an N-dimensional vector. Then, we pass the output through a softmax function. Each entry corresponds to a bin and represents the probability that the ground truth falls within the corresponding interval.
3. After obtaining the probability distribution of length N from the model, we calculate the expected value of this distribution, which serves as the predicted result for the regression task.

The dual loss L_{dual} can be obtained after modifying the model. As shown in Fig. 4, in addition to computing the error of the output values in the regression task (L_{reg}), we also computed the classification error of the generated probability distribution (L_{cls}). The corresponding ground truth is represented as a one-hot vector, where the bin corresponding to the ground truth is assigned a value of 1, and the rest is assigned a value of 0. The dual loss is then defined as follows:

$$L_{dual} = L_{reg} + a * L_{cls} \tag{1}$$

where $\alpha \in R$ is a hyperparameter that controls the weight between the two losses, we used the common Mean Squared Error (MSE) as L_{reg}. As for L_{cls}, it is crucial for dealing with imbalanced data. Since imbalance issues in classification tasks have been well studied as compared to those in regression tasks, methods such as SMOTEchawla2002smote or Focal Loss [21] have been widely used. By introducing the

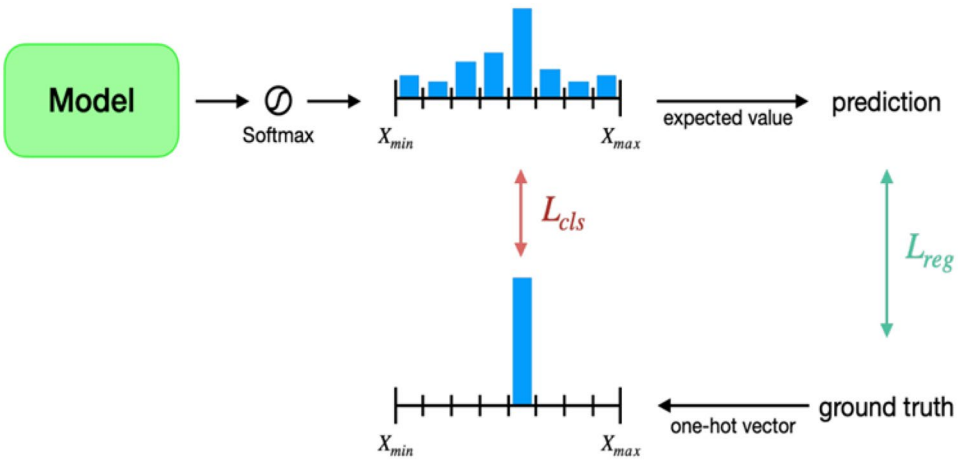


Fig. 4 Dual loss

classification-like mechanism to our task, we can use these approaches to handle the imbalance problem in the data.

Results

In this section, the effectiveness of the model was demonstrated through various experimental results. We mainly conducted experiments on the EYES-DEFY-ANEMIA dataset [8] and the National Taiwan University Hospital (NTUH) dataset, evaluating the performance of both regression and classification tasks. Additionally, we performed an ablation study to investigate the impact of different modules on the accuracy of the model. Furthermore, with the growing importance of Explainable AI in recent years, interpretability becomes particularly crucial for prediction tasks in the medical field. Therefore, we presented different qualitative results to explain the behavior of the model at different stages. We computed p values for the differences in performance (mean absolute error [MAE] or F1 score) under different scenarios using the 1-tailed permutation test with 1000 iterations. A p value less than 0.05 was considered statistically significant.

Implementation Details

In our experiments, we used the PyTorch [22] framework for implementation. We performed computations on an NVIDIA RTX 3090 GPU to accelerate the training process. We used the AdamW [23] optimizer with a learning rate of 10^{-4} to optimize the model parameters during training. The batch size is set to 16 to balance computational efficiency and memory usage. We used two datasets: EYES-DEFY-ANEMIA and NTUH dataset. We trained and validated our model on EYES-DEFY-ANEMIA dataset by using 5-fold cross validation with 140 epochs. Moreover, we retrained our model on NTUH dataset by using 4-fold cross validation with 150 epochs.

As for the detailed setup of BPANet, in the data preprocessing stage, we used 2 iterations of the opening operation to remove the reflective noise, and then resize the image to 224*224 to match the input dimension of ResNet. In the backbone model, we used ResNet50 and add Channel-Spatial Attention Module between Stage1 and Stage2, and the output of this stage is a feature vector of length 1000. Meanwhile, we converted Gender to a binary value of 0 and 1, Age is divided by 100 to ensure that the value falls between 0 and 1, and the Body Part Tag is encoded into a three-dimensional one-hot vector. Finally, in the Dual Loss Module, we set the minimum and maximum values of the output range at [0, 24], and cut it into 48 bins, we took MSE as the loss function of the regression part, and used Focal Loss [21] as the criteria of the classification part to handle the data imbalance problem, and the weighting hyperparameter α is set to 0.5.

Data Sets

1. Retrospective: EYES-DEFY-ANEMIA dataset. Regarding the nature of the body part images, the eye images we used are comparatively more complex than the other three body parts. Therefore, we evaluated the performance of the model on eye images using the publicly available EYES-DEFY-ANEMIA dataset [8]. This dataset collects images from 218 patients in Italy and India. The full-eye photos of each patient were captured by a smartphone combined with a device for stabilizing environmental factors. In addition to the images, the dataset also contains information such as gender, age, and corresponding hemoglobin concentrations for each patient, which serve as ground truth for our regression task.

The data distribution of regression and classification tasks in the dataset are shown in Figs. 5 and 6, respectively. The labels for the classification task are according to the WHO-defined thresholds of 13 g/dL for males and 12 g/dL for females. Last but not least, the variables of gender and age, which serve as part of the inputs to the model, also affect the prediction significantly. More details will be discussed in the following sections.

2. Prospective: NTUH dataset. We prospectively collected data from January, 2023 to November, 2023 in the emergency department (ED) of the NTUH in Taiwan. Well-trained research personnel successively enrolled patients who visited the ED following a standard protocol. The inclusion criteria were age of 18 years or above, had a hemoglobin (Hb) measurement in the ED, and the ability to give informed consent. The time interval between blood draw and image capture was controlled within 12 h to ensure the accuracy of the ground truth.

We excluded patients who had active eye conditions, experienced hypoxemia (SpO_2<90%), or received transfusion between the time of Hb measurement and the time photos were taken. The NTUH Institutional Review Board approved this study, and informed consent was obtained for all participants. A total of 101 patients were included in our study. We used an iPhone 13 (Apple Inc., CA, USA) to capture the images of the patient's conjunctiva, palm, and fingernails.

The triage data were also collected, including basic demographics, mode of arrival, past medical history, triage level, chief complaint, and disposition. To simulate real-world scenarios, we ensured an adequate light source (ambient

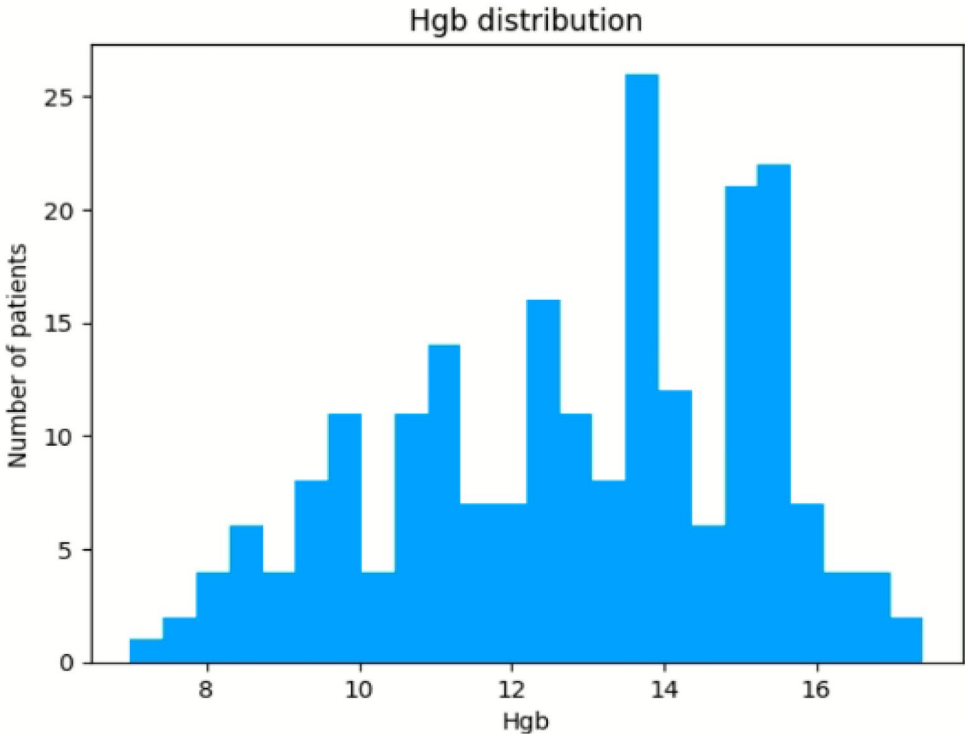


Fig. 5 Distribution of hemoglobin data in EYES-DEFY-ANEMIA

lighting) during image acquisition and did not impose excessive restrictions on the shooting angles. Then, we manually selected appropriate regions of interest (ROI) as image data. In addition to the images, patient demographics, including gender and age, are also recorded.

Finally, the gold standard for anemia diagnosis, hemoglobin concentration (Hgb), was obtained by blood sampling. Unlike the EYES-DEFY-ANEMIA dataset where generally healthy subjects were included, the NTUH data were more diverse because the ED had a diverse set of patients with co-morbid conditions.

Table 1 showed the demographics of the 101 ED patients. The mean age was 67.1 years, and 43% were men. Figures 7 and 8 show that the distribution of Hgb values of

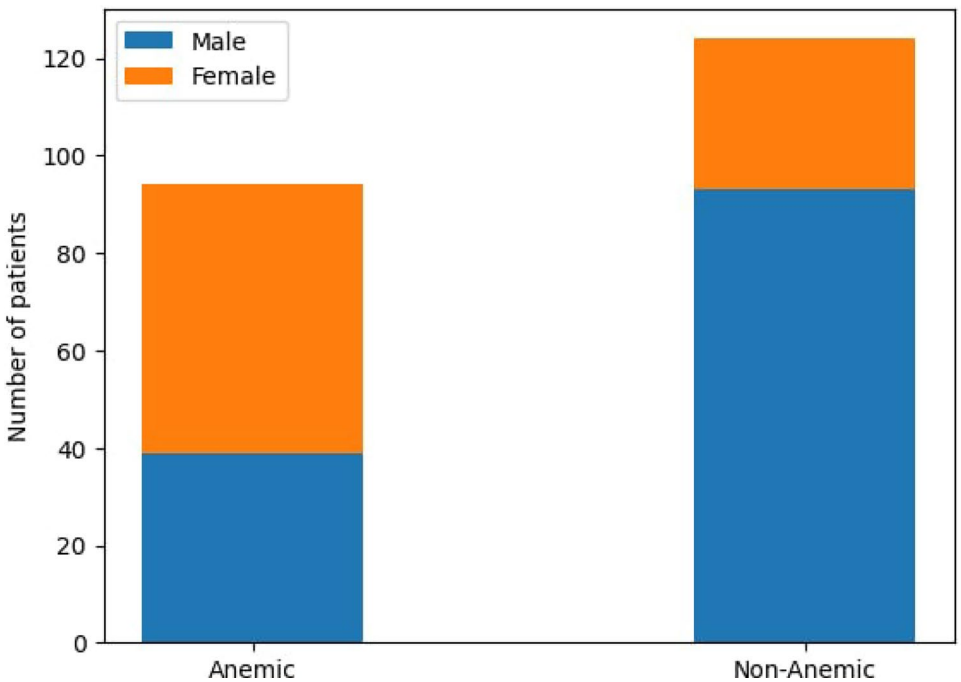


Fig. 6 Distribution of classification data in EYES-DEFY-ANEMIA

Table 1 Baseline clinical characteristics of emergency department patients in National Taiwan University hospital

Variable	N = 101			
	Statistic	Value	Statistic	Value
Age, yr	mean	67.1	SD	16.9
Female sex	count	57	%	56.4
Arrival by ambulance	count	6	%	5.9
Most common chief complaint				
Fever	count	17	%	16.8
Abdominal pain	count	12	%	11.9
Shortness of breath	count	10	%	9.9
Triage level				
1 (Resuscitation)	count	1	%	1.0
2 (Emergent)	count	15	%	14.9
3 (Urgent)	count	85	%	84.2
4 (Less urgent)	count	0	%	0
5 (Non-urgent)	count	0	%	0
Anemia	count	58	%	57.4
Hb level, g/dL	mean	11.3	SD	2.8
Hospital admission	count	61	%	60.4
ED length of stay, hr	median	45.8	IQR	26.6–70.3

SD standard deviation, *IQR* interquartile range, *ED* emergency department

the 101 patients. Anemia was defined as a Hb value < 13.1 g/dL for men, and < 12 g/dL for women according to the NTUH laboratory. Fifty-seven percent of the patients were in the anemic range. In addition, the emergency department environment introduces more interference factors, such as unfixed light sources and shadows during image acquisition. Therefore, it was necessary to properly remove data with excessive noise and perform denoising before feeding

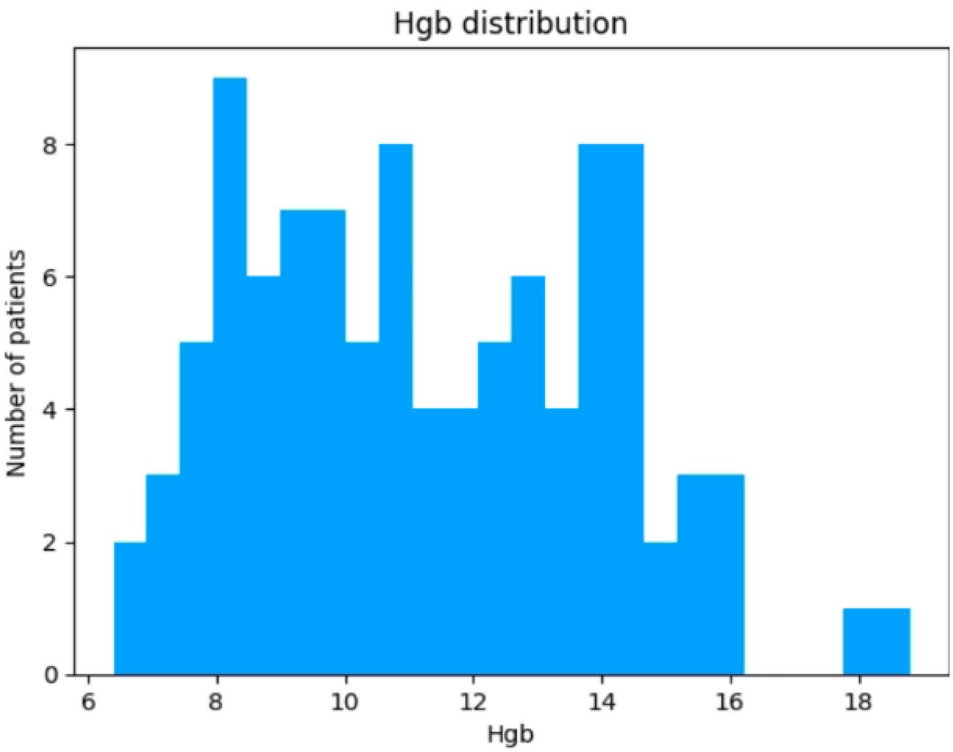


Fig. 7 Distribution of hemoglobin data in National Taiwan University Hospital dataset

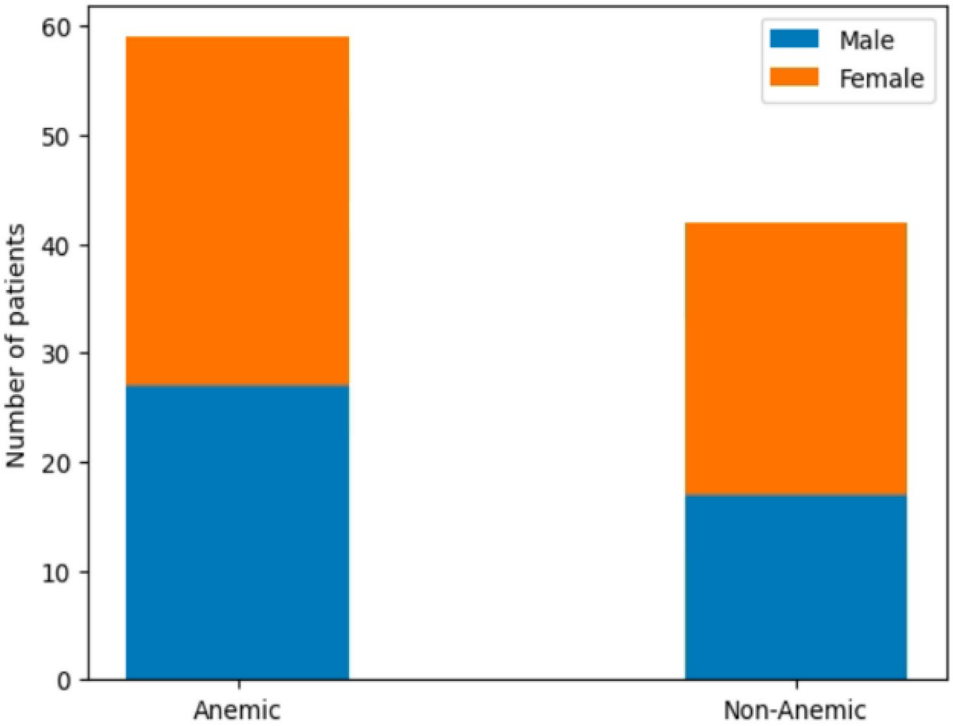


Fig. 8 Classification of hemoglobin data in National Taiwan University Hospital dataset

the photos into the model. After the screening process, 100/85/83 images from conjunctiva, palm and fingernail were retained, respectively ($n = 268$ in total).**Fig. 7** Distribution of hemoglobin data in National Taiwan University Hospital dataset**Fig. 8** Classification of hemoglobin data in National Taiwan University Hospital dataset

Evaluation Metrics

Our experiments primarily focused on the regression task of hemoglobin prediction and the derived classification task of anemia detection. In the former task, we used Mean Absolute Error (MAE) as a metric to evaluate the regression errors. This metric reflects the average error of the model in predicting hemoglobin concentration and provides an intuitive measure of the model's predictive performance in the regression task.

For the anemia detection task, the model first predicted the hemoglobin values and then classifies whether the individual has anemia or not based on the threshold defined by the WHO or the NTUH, depending on the dataset. In terms of performance evaluation, we considered anemic patients as positive samples and non-anemic individuals as negative samples and used various metrics calculated from the confusion matrix.

We used accuracy (Acc) to measure the overall correctness of predictions, while recall focuses on correctly identifying true anemia cases. Precision measures the proportion of true anemia cases among all predicted anemia cases. F1-score provides a balanced evaluation by combining precision and recall into a single measure. These metrics collectively assess the model's ability to accurately detect anemia and strike a balance between false positives and false negatives.

Results on EYES-DEFY-ANEMIA Dataset

1. *Quantitative Results:* Since the EYES-DEFY-ANEMIA dataset is relatively new, the only comparable results currently available are from the original study conducted by Dimauro et al. [8]. They used manually designed features combined with the RUSBoost machine learning method for prediction. Although the dataset includes the ground truth values of Hgb, the original study focused only on the classification task and reported metrics such as accuracy, recall, and specificity. Using these three metrics along with the distribution of positive and negative samples in the dataset, we derived the corresponding confusion matrix metrics. After fivefold cross-validation, the results, shown in Table 2, demonstrate that our model outperformed their method.
2. *Qualitative Results:* The Channel-Spatial Attention (CSA) module had a significant impact on the extraction

Table 2 A comparison with previous method on the EYES-DEFY-ANEMIA dataset. (Unit: g/dL)

Method	Metrics				
	MAE	Acc	Recall	Precision	F1-score
Dimauro et al. * MERGEFORMAT [8]	-	0.83	0.66	0.56	0.61
BPANet (Ours)	**1.212**	**0.849**	**0.876**	**0.798**	**0.828**

MAE mean absolute error, *Acc* accuracy

of conjunctival features. We demonstrated the benefits of introducing this module through visualizations as shown in Fig. 9. For each input image, we presented the output feature maps of CSA. In addition, we showed the output results of stage 4 to compare the effect of this module on the feature extraction capability of the backbone model. It can be observed that after the introduction of CSA, the model was able to effectively identify the region of interest corresponding to the conjunctiva from eye images, and thus focused on feature extraction within this specific area.

3. *Experimental Analysis of Dual Loss Module:* Finally, we conducted experiments on the relevant hyperparameters of the dual loss module. The main purpose of the dual loss module in our model is to address the issue of data imbalance. Therefore, we conducted the

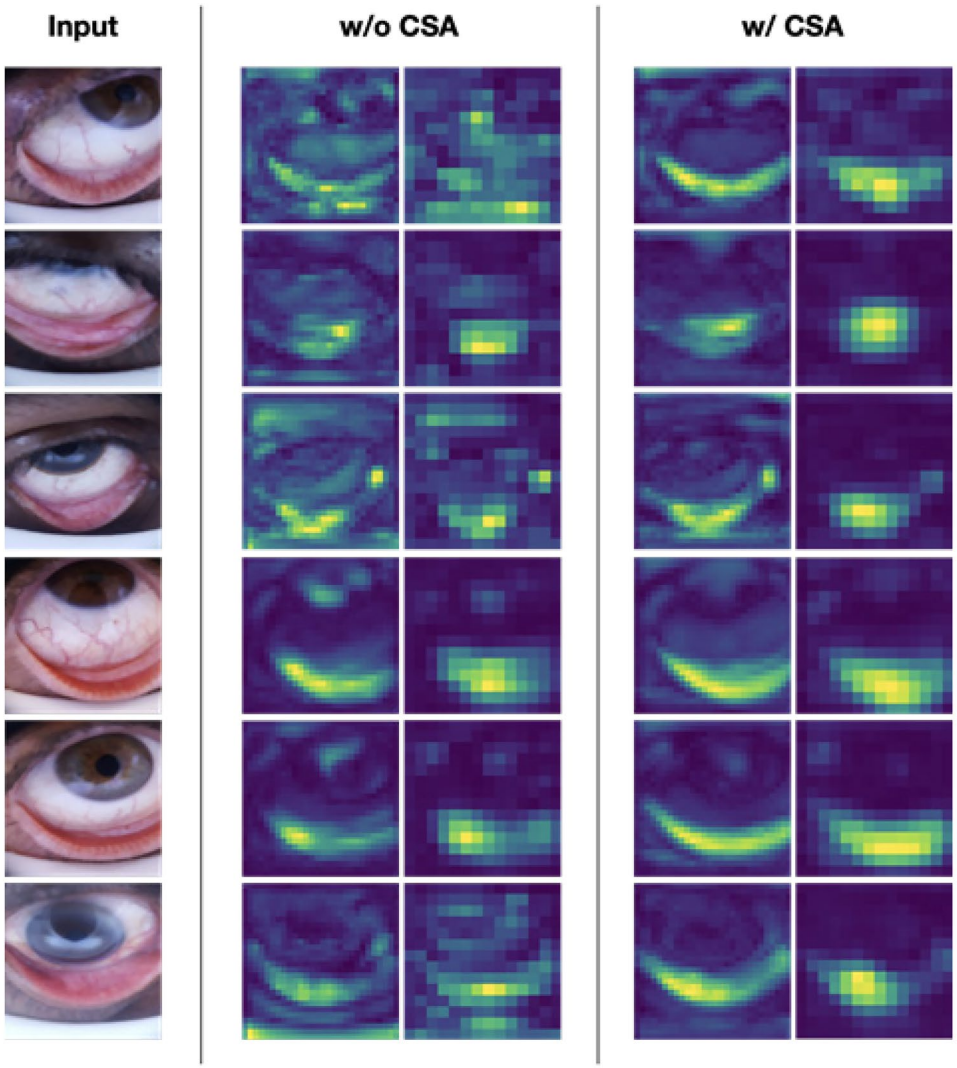


Fig. 9 Comparison of using the CSA module. The central column shows the result without CSA, while the right column shows the result with CSA

Table 3 Experimental results for hyperparameter ***N*** in the dual loss module. (Unit: g/dL)

N	Metrics					
	MAE	*P* value for pairwise comparison in MAE	Acc	Precision	F1-score	*P* value for pairwise comparison in F1-scoree
24	1.303	0.491	0.811	0.792	0.774	0.084
48	**1.212**	**Reference**	**0.849**	**0.798**	**0.828**	**Reference**
72	1.27	0.894	0.784	0.715	0.758	0.037

MAE mean absolute error, *Acc* accuracy

experiments on the EYES-DEFY-ANEMIA dataset, which has a more severe imbalance problem. In this dataset, anemic patients account for only 20% of the total, and the hemoglobin values are concentrated in the range of 13 to 15. To allow the model to handle different extreme data cases, we set the X_{min} and X_{max} parameters in the module to 0 and 24, respectively. We first conducted experiments on the hyperparameter N, which represents the length of the probability distribution produced by the model. N also affects the size of each bin, with smaller N resulting in larger bin sizes. A smaller N indicates that the model is more likely to reduce the classification loss (L_{cls}), but it also means that the model learns less information from L_{cls}. On the other hand, if N is too large, the classification part becomes similar to the regression part, which causes the model to lose its ability to handle the data imbalance effectively. The experimental results are shown in Table 3. Based on these results, we eventually set N to 48.

Results on NTUH Dataset

1) Quantitative Results: We evaluated the performance of BPANet using fourfold cross-validation, with each body part evenly distributed across different folds. The results are shown in Table 4. Since the NTUH dataset was a private dataset, and to the best of our knowledge, there is no existing research supporting the use of multiple body parts for prediction. Therefore, we did not have a comparative benchmark for this aspect. To validate the effectiveness of the model, we followed the recommendation of medical experts and used a clinically simplified method [24] to evaluate the accuracy of the regression task. This method first obtains the maximum (X_{max}) and minimum (X_{min}) values from the ground truth of the dataset. If the Mean Absolute Error (MAE) of the target model is less than $(X_{max} - X_{min})/4$, it can be considered to have an acceptable level of accuracy. In the case of the NTUH dataset, $[X_{max}, X_{min}] = [17.8, 6.4]$. Therefore, the MAE of the model should be less than $(17.8 - 6.4)/4 = 2.85$. The results show that our designed BPANet indeed achieves such accuracy within these criteria.

Table 4 Quantitative results on the National Taiwan University Hospital dataset. (Unit: g/dL)

Method	Metrics				
	MAE	Acc	Recall	Precision	F1-score
BPANet	**2.024**	**0.716**	**0.857**	**0.718**	**0.788**

MAE mean absolute error, *Acc* accuracy

We also analyzed the influence of different body parts on the performance of BPANet. The experiments are also conducted using 4-fold cross-validation but with a difference in the evaluation process. For each validation fold, we computed the Mean Absolute Error (MAE) and classification metrics for the target body part and observe their influence on the model's predictions.

As shown in Table 5, based on the model's performance on different body parts, the ranking from high to low can be summarized as follows: Palm > Conjunctiva > Fingernail. First, the best-performing data came from the palm region. This region is commonly used in physical examinations and contains abundant hidden information that allows the model to make stable predictions. The second-best performance was observed in the conjunctiva data. In addition to the eye crust, external factors such as eyelashes may introduce further interference and affect the performance of the model. Finally, the fingernail images showed the worst performance.

Last, we investigated the effect of data volume for each body part in the dataset on model training. The experiment was performed using 4-fold cross-validation, where the validation set remained unchanged, but we reduced the data volume of the target body part in the training set by 20%. The experimental results are presented in Table 6.

The results demonstrate similar findings as the previous experiment, and we attributed these results to the body part

Table 5 Comparison of model performance across different body parts. (Unit: g/dL)

Body part	Metrics						
	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
Conjunctiva	2.006	0.865	0.720	0.861	0.717	0.781	0.807
Palm	**1.844**	0.309	**0.800**	**0.918**	**0.783**	**0.841**	0.123
Fingernail	2.232	0.223	0.625	0.792	0.657	0.707	0.276
BPANet (Ours)	2.024	**Reference**	0.716	0.857	0.718	0.788	**Reference**

MAE mean absolute error, *Acc* accuracy

Table 6 Effect of training data with different body parts on model performance. (Unit: g/dL)

Target body part	Metrics						
	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
Conjunctiva	2.266	0.022	0.659	0.691	0.717	0.706	0.132
Palm	2.080	0.731	**0.729**	0.758	**0.806**	0.765	0.337
Fingernail	2.353	0.238	0.612	0.689	0.671	0.669	0.226
BPANet (Ours)	**2.024**	**Reference**	0.716	**0.857**	0.718	**0.788**	**Reference**

MAE Mean absolute error, Acc accuracy

Table 7 Ablation study results of various modules in BPANet on the EYES-DEFY-ANEMIA dataset. (Unit: g/dL)

Modules			Metrics						
CSA	FA	DL	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
-	-	-	1.479	0.045	0.770	0.836	0.715	0.749	0.029
-	-	✓	1.308	0.908	0.788	0.802	0.740	0.760	0.006
-	✓	-	1.337	0.775	0.816	0.835	0.771	0.792	0.084
✓	-	-	1.320	0.614	0.784	0.840	0.709	0.792	0.078
✓	✓	-	1.282	0.871	0.816	0.853	0.747	0.762	0.028
✓	-	✓	1.282	0.697	0.829	0.833	0.784	0.804	0.110
-	✓	✓	1.286	0.633	0.840	0.817	**0.819**	0.810	0.257
✓	✓	✓	**1.212**	**Reference**	**0.849**	**0.876**	0.798	**0.828**	**Reference**

CSA Channel-Spatial attention, *FA* Fusion Attention, *DL* Dual Loss, *MAE* mean absolute error, *Acc*, accuracy

Table 8 Ablation study results of different inputs of BPANet on the EYES-DEFY-ANEMIA dataset. (Unit: g/dL)

Input		Metrics						
Age	Gender	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
-	-	1.460	0.605	0.784	0.782	0.752	0.748	0.164
✓	-	1.463	0.931	0.802	0.865	0.763	0.783	0.003
-	✓	1.395	0.931	0.811	0.790	0.773	0.776	0.018
✓	✓	**1.121**	**Reference**	**0.849**	**0.876**	**0.798**	**0.828**	**Reference**

MAE mean absolute error, *Acc* accuracy

Table 9 Ablation study results of various modules in BPANet on the National Taiwan University Hospital dataset. (Unit: g/dL)

Modules			Metrics						
CSA	FA	DL	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
-	-	-	2.293	0.067	0.616	0.626	0.693	0.655	0.034
-	-	✓	2.193	0.130	0.680	0.700	**0.745**	0.708	0.049
-	✓	-	2.155	0.298	0.642	0.716	0.687	0.699	0.280
✓	-	-	2.212	0.212	0.628	0.697	0.678	0.685	0.117
✓	✓	-	2.133	0.481	0.687	0.830	0.698	0.756	0.488
✓	-	✓	2.175	0.186	0.681	0.742	0.721	0.729	0.685
-	✓	✓	2.161	0.246	0.694	0.739	0.739	0.739	0.155
✓	✓	✓	**2.024**	**Reference**	**0.716**	**0.857**	0.718	**0.788**	**Reference**

CSA Channel-Spatial attention, *FA* Fusion Attention, *DL* Dual Loss, *MAE* mean absolute error, *Acc* accuracy

Table 10 Ablation study results of different inputs of BPANet on the National Taiwan University Hospital dataset. (Unit: g/dL)

Input		Metrics						
Age	Gender	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
-	-	2.271	0.048	0.643	0.677	0.702	0.689	0.029
✓	-	2.245	0.056	0.623	0.690	0.677	0.681	0.011
-	✓	2.218	0.153	0.650	0.734	0.689	0.710	0.089
✓	✓	**2.024**	**Reference**	**0.716**	**0.857**	**0.718**	**0.788**	**Reference**

MAE mean absolute error, *Acc* accuracy

characteristics, emphasizing the palm yields the greatest improvement in model accuracy. After the palm, the conjunctiva and fingernail also show a slight performance improvement. It is worth noting that the model benefits from augmentation regardless of the target body part, indicating that our model can still benefit from increasing training data to achieve further improvements.

Ablation Study

1. *Results on EYES-DEFY-ANEMIA Dataset:* Table 7 shows the results of the ablation study performed on the EYES-DEFY-ANEMIA dataset. We compared the impact of the channel-spatial attention module (CSA), fusion attention module (FA), and dual loss module (DL) on the performance

Table 11 A comparison of using body part tags on the prediction performance. (Unit: g/dL)

Method	Metrics						
	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
w/o Body Part Tag	2.152	0.041	0.662	0.73	0.700	0.714	0.021
w/ Body Part Tag	**2.024**	**Reference**	**0.716**	**0.857**	**0.718**	**0.788**	**Reference**

MAE mean absolute error, *Acc* accuracy

Table 12 Effect of data augmentation on the EYES-DEFY-ANEMIA dataset. (Unit: g/dL)

Method	Metrics						
	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
w/ Augmentation	1.232	0.350	0.826	0.871	0.771	0.813	0.370
w/o Augmentation	**1.212**	**Reference**	**0.849**	**0.876**	**0.798**	**0.828**	**Reference**

MAE mean absolute error, *Acc* accuracy

Table 13 Effect of data augmentation on the National Taiwan University Hospital dataset. (Unit: g/dL)

Method	Metrics						
	MAE	*P* value for pairwise comparison in MAE	Acc	Recall	Precision	F1-score	*P* value for pairwise comparison in F1-score
w/ Augmentation	2.125	0.170	0.710	0.838	0.723	0.778	0.440
w/o Augmentation	**2.024**	**Reference**	**0.716**	**0.857**	**0.718**	**0.788**	**Reference**

MAE mean absolute error, *Acc* accuracy

Table 14 Comparison between some related studies and our proposed work

Paper	Methods	Using demographic variables as part of the features (e.g., age, gender)	Flexibility using multiple body parts	Dual datasets (retrospective + prospective)
Zhang et al. [13]	InceptionV3 + Face + +	No	No	No
Zheng et al. [14]	ROI + Two-Branch EfficientNet	No	No	No
Appiahene et al. [9]	ROI + CIELAB + CNN + k-NN + Naive Bayes	No	No	No
Ours	ResNet50 + CSA + FA + DL	**Yes**	**Yes**	**Yes**

ROI region of interest, *CIELAB* CIE L*a*b*, *CSA* Channel-Spatial attention, *FA* Fusion Attention, *DL* Dual Loss

of the model. The CSA and FA modules can be easily removed due to their design, while in the experiment without DA, the model directly predicted the values instead of using indirect prediction. Several conclusions can be drawn from the experimental data. First, since this dataset contained only eye images, the importance of FA was relatively lower compared to datasets with multiple body parts, such as the NTUH dataset. However, FA still played an important role in feature fusion between image features and demographic variables. Second, due to the significant data imbalance in this dataset, removing DL led to a noticeable decrease in model performance due to bias. Finally, the spatial attention module in CSA was particularly critical for eye data, and thus CSA had a strong impact on the performance of this dataset due to its ability in feature extraction. The comparisons of all vs. few/none modules were generally statistically significant.

Table 8 demonstrates the impact of gender and age on the prediction task. It is worth noting that since this dataset contains only a single body part, we did not perform experiments with body part tags. From the results in the table, we can observe that both gender and age have a significant impact on the model's predictions due to their strong correlation with hemoglobin levels.

2. *Results on NTUH Dataset:* We also conducted experiments on the NTUH dataset to validate the performance of these modules on a dataset containing multiple body part images. As mentioned in the previous section, the overall performance decreased slightly on this dataset due to its complexity and noise. As shown in Table 9, in contrast to the EYES-DEFY-ANEMIA dataset, the importance of CSA and FA in improving feature extraction was highlighted, as the input images are no longer singular. Furthermore, the imbalance problem was less pronounced in this dataset, resulting in a less prominent advantage of DL. However, DL still contributed to the convergence of the model and leads to a slight performance improvement. The comparisons of all vs. few/none modules were in general statistically significant.

Next, we conducted an analysis of the non-image inputs. From the results in Table 10, we obtained similar conclusions to those of the EYES-DEFY-ANEMIA dataset, further confirming the importance of gender and age in predicting hemoglobin levels. Table 11 shows the experimental result of our main contribution, the body part tag. The data clearly showed a statistically significant improvement in performance after incorporating body part tags, especially in more challenging regression tasks. This improvement was attributed to the fact that the model no longer treats all input images as the same data but instead leveraged the fusion attention module to enhance relevant features based on the body part tags. These results validated the need for incorporating body part tags into the design.

Data Augmentation

Finally, we experimented with one common data augmentation technique: random rotation. We applied random rotations, either clockwise or counterclockwise by 15 degrees, to the images in both datasets. The effects of this augmentation are shown in Table 12 for the EYES-DEFY-ANEMIA dataset and in Table 13 for the NTUH dataset. As evident from these tables, the data augmentation did not yield statistically significant improvements in model performance on either dataset.

Discussion

In this paper, we proposed a deep learning model, BPANet, that enables flexible prediction of hemoglobin concentration (Hgb) using images of different body parts. Our model is novel in several aspects. First, by incorporating the body part tag and demographic variables such as gender and age, the model leverages attention mechanisms to highlight relevant image features, thereby achieving accurate Hgb prediction and facilitating anemia detection. Second, we employed the channel-spatial attention mechanism to enable direct prediction of the conjunctiva image without the need for complicated image segmentation processes, thereby enhancing the feature extraction capability of the backbone model. We also provided interpretability through heat map analysis, shedding light on the model's attention process. Third, with the dual loss design, our model not only addressed the problem of data imbalance but also utilized the characteristics of Hgb to accelerate model training and improve accuracy.

Our BPANet model significantly outperformed Dimauro et al. [8] method using the same EYES-DEFY-ANEMIA dataset. The accuracy increased from 0.830 to 0.849, and the F1-score from 0.61 to 0.828. We attributed the improvement made to several factors. First, by utilizing ResNet50 as the backbone model, BPANet autonomously extracts crucial features without relying on manually designed features. This reduced the potential bias introduced by subjective human factors. The improvement can be confirmed by the results of the ablation study, which demonstrate the significant impact of this change alone on performance. Second, different attention mechanism modules not only improve the feature extraction capability, but also enhance the interpretability of the model. Finally, the dual loss function allows the model to make predictions within the specified range, thereby improving both training speed and accuracy.

To compare with previous machine and deep learning-based studies, we included Abbreviation: *MAE*, mean absolute error; *Acc*, accuracy.

Table 14, summarizing key differences between our approach and theirs. Unlike models that are limited to a single body part, BPANet introduces flexibility by accepting images from various body parts, including conjunctiva, nails, and palms. This flexibility is crucial in real-world applications because the availability of specific body part images may be constrained due to patient conditions or other factors. Inputting basic demographics is also a simple addition to applications that may improve model performance. Using both retrospective and prospective datasets also enhances our model's generalizability to the real-world setting.

To re-evaluate the effectiveness of our model, we presented a series of experimental results on a separate NTUH dataset. As expected, the overall performance decreased slightly on this dataset due to the inclusion of older patients with multiple comorbidities and real-world emergency department settings. Nonetheless, the accuracy and F1-score of the model can still maintain at 0.716 and 0.788, respectively. We also found that palm region may best serve as the ROI when developing future mobile applications because of the better model performance using this region. The ablation studies confirmed the important of each module component, although the emphases were slightly different due to different data composition.

Many studies, such as those by Jain et al. [25], Appiahene et al. [9], Asare et al. [10], Magdalena et al. [26], and Agrawal [27], have mentioned that avoiding over-fitting and under-fitting in machine and deep learning models typically requires a large dataset. Although our research utilized relatively smaller datasets, we did not apply data augmentation to the training set to alleviate the scarcity of data. This decision was primarily due to the nature of hemoglobin detection, which relies heavily on color accuracy. Most data augmentation techniques involve adjustments such as color modification and resizing, which could potentially distort the critical features related to hemoglobin levels, leading to inaccurate results. Additionally, cropping could inadvertently remove essential regions of interest, compromising the integrity of the analysis. Given these considerations, standard data augmentation methods may not be suitable for this task, as they could introduce variability that may undermine the reliability of the detection process. To explore the impact of data augmentation, we also performed additional experiments which showed no obvious improvement in model performance.

Instead, we employed Neural Architecture Search (NAS) to select models that are robust against over-fitting. NAS helps in identifying the optimal architecture that generalizes well across the dataset. Additionally, we used k-fold cross-validation to rigorously test our models and detect any signs of over-fitting. Through this process, we did not observe over-fitting in our results. Furthermore, our dataset was relatively balanced, which likely contributed to the absence of over-fitting. A balanced dataset helps the model learn the underlying patterns more effectively without biasing towards any particular class.

Despite the promising results, our study has several limitations. First, the model's performance was dependent on the quality and variety of the training data. While we included multiple body parts and diverse images, further external validation studies are needed to evaluate the model's robustness in different populations and settings. Second, the reliance on image-based data may introduce biases due to variations in lighting and image capture conditions. Future work should focus on enhancing the model's robustness to such variations. Third, while our approach minimized preprocessing, it still required initial image capture conditions to be consistent. Investigating more advanced techniques for real-time image normalization could further improve the model's applicability in varied settings. Finally, the deployment of the model on real-time platforms was not tested, which is a potential area for future research to ensure the model's applicability.

Conclusion

This paper presents a novel approach for predicting hemoglobin levels that provides flexibility by using multiple body part images for prediction. This approach holds potential for home anemia screening using mobile applications. BPANet requires only minor additions to the user interface and can provide users with real-time hemoglobin predictions. This quick home screening may inform patients of the need for additional hospital visit. In the emergency department, quick anemia detection can aid inexperienced physicians and speed up the blood preparation, potentially saving more lives.

Author Contribution All authors contributed to the study conception and design. Data collection was performed by Chia-Hsin Ko. Data preprocessing and analysis were performed by En-Ting Lin and Shao-Chi Lu. The first draft of the manuscript was written by En-Ting Lin and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript. Drs. Chu-Lin Tsai and Li-Chen Fu supervised the project.

Funding This research was supported by the Ministry of Science and Technology of Taiwan, and Center for Artificial Intelligence & Advanced Robotics, National Taiwan University, under the grant numbers MOST 110–2634-F-002–049 & MOST 110- 2221-E-002–166-MY3. This project was also supported by grants from the National Health Research Institutes (NHRI-EX113-11332PI), the National Science and Technology Council (NSTC 112–2314-B-002–264) and the National Taiwan University Hospital (112-UN-0027 and 113-UN0017).

Data Availability The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Declarations

Ethics Approval The National Taiwan University Hospital (NTUH) Institutional Review Board approved this study (202210071RIND), and informed consent was obtained for all participants.

Consent to Participate Informed consent was obtained from all participants in the National Taiwan University Hospital study.

Consent for Publication Informed consent was obtained from all participants in the National Taiwan University Hospital study for analyzing and publish their data. The manuscript does not contain any individual person's images or videos from the National Taiwan University Hospital study.

Competing Interests The authors declare no competing interests.

References

1. R. J. Stoltzfus, A. Edward-Raj, M. L. Dreyfuss, *et al.*: Clinical pallor is useful to detect severe anemia in populations where anemia is prevalent and severe. *The Journal of nutrition*, vol. 129, no. 9, pp. 1675–1681, 1999.
2. S. Suner, G. Crawford, J. McMurdy, and G. Jay: Non-invasive determination of hemoglobin by digital photography of palpebral conjunctiva. *The Journal of emergency medicine*, vol. 33, no. 2, pp. 105–111, 2007.
3. A. Tamir, C. S. Jahan, M. S. Saif, et al.: Detection of anemia from image of the anterior conjunctiva of the eye by image processing and thresholding. in *2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC)*, IEEE, 2017, pp. 697–701.
4. M. I. Khan, R. K. Mondol, M. A. Zamee and T. A. Tarique, "Hardware architecture design of Anemia detecting regression model based on FPGA," 2014 International Conference on Informatics, Electronics & Vision (ICIEV), Dhaka, Bangladesh, 2014, pp. 1–5, https://doi.org/10.1109/ICIEV.2014.6850814.
5. S. Collings, O. Thompson, E. Hirst, L. Goossens, A. George, and R. Weinkove: Non-invasive detection of anemia using digital photographs of the conjunctiva. *PloS one*, vol. 11, no. 4, e0153286, 2016.
6. Y.-M. Chen and S.-G. Miaou: A Kalman filtering and nonlinear penalty regression approach for noninvasive anemia detection with palpebral conjunctiva images. *Journal of healthcare engineering*, vol. 2017, 2017.
7. M. Anggraeni and A. Fatoni: Non-invasive self-care anemia detection during pregnancy using a smartphone camera. in *IOP Conference Series: Materials Science and Engineering, IOP Publishing*, vol. 172, 2017, p. 012 030.
8. G. Dimauro, M. E. Griseta, M. G. Camporeale, F. Clemente, A. Guarini, and R. Maglietta: An intelligent non-invasive system for automated diagnosis of anemia exploiting a novel dataset. *Artificial Intelligence in Medicine*, vol. 136, p. 102 477, 2023.
9. P. Appiahene, J. W. Asare, E. T. Donkoh, G. Dimauro, and R. Maglietta: Detection of iron deficiency anemia by medical images: A comparative study of machine learning algorithms. *BioData Mining*, vol. 16, no. 1, pp. 1–20, 2023.
10. J. W. Asare, P. Appiahene, E. T. Donkoh, and G. Dimauro, "Iron deficiency anemia detection using machine learning models: A comparative study of fingernails, palm and conjunctiva of the eye images," Engineering Reports, e12667, 2023.
11. E. J. Wang, W. Li, D. Hawkins, T. Gernsheimer, C. Norby-Slycord, and S. N. Patel: Hemaapp: Noninvasive blood screening of hemoglobin using smartphone cameras. in *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 2016, pp. 593–604.
12. R. G. Mannino, D. R. Myers, E. A. Tyburski, et al.: Smartphone app for non-invasive detection of anemia using only patient-sourced photos. *Nature communications*, vol. 9, no. 1, p. 4924, 2018.
13. A. Zhang, J. Lou, Z. Pan, et al.: Prediction of anemia using facial images and deep learning technology in the emergency department. *Frontiers in Public Health*, vol. 10, p. 3917, 2022.
14. L. Zheng, S. Liu, S. Tian, et al.: Enhancing intelligent anemia detection via unifying global and local views of conjunctiva image with two-branch neural networks. 2022.
15. S. Roychowdhury, P. Hage, and J. Vasquez: Azure-based smart monitoring system for anemia-like pallor. *Future Internet*, vol. 9, no. 3, p. 39, 2017.
16. A. Kesarwani, S. Das, M. Dalui, et al.: Non-invasive anemia detection by examining palm pallor: A smartphone-based approach. *Biomedical Signal Processing and Control*, vol. 79, p. 104 045, 2023.
17. M. E. Salive, J. Cornoni-Huntley, J. M. Guralnik, *et al.*: Anemia and hemoglobin levels in older persons: Relationship with age, gender, and health status. *Journal of the American Geriatrics Society*, vol. 40, no. 5, pp. 489–496, 1992.
18. S. Kasiviswanathan, T. Bai Vijayan, L. Simone, and G. Di- mauro: Semantic segmentation of conjunctiva region for non- invasive anemia detection applications. *Electronics*, vol. 9, no. 8, p. 1309, 2020.
19. S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon: Cbam: Convolutional block attention module. in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 3–19.
20. M.-H. Guo, Z.-N. Liu, T.-J. Mu, and S.-M. Hu: Beyond self- attention: External attention using two linear layers for visual tasks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.
21. T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dolla´r: Focal loss for dense object detection. in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2980– 2988.
22. A. Paszke, S. Gross, F. Massa, *et al.*: Pytorch: An imperative style, high-performance deep learning library. in *Advances in Neural Information Processing Systems 32*, Curran Associates, Inc., 2019, pp. 8024–8035. [Online]. Available: http://papers.neurips.cc/paper/9015-pytorch-an-imperative-style-high-performance-deep-learning-library.pdf.
23. I. Loshchilov and F. Hutter: Decoupled weight decay regularization. *arXiv preprint* arXiv:1711.05101, *2017*.
24. Watt, J.A., Veroniki, A.A., Tricco, A.C. et al.: Using a distribution-based approach and systematic review methods to derive minimum clinically important differences. BMC Med Res Methodol 21, 41 (2021)
25. Jain, Prakhar, Shubham Bauskar, and Manasi Gyanchandani.: "Neural network based non-invasive method to detect anemia from images of eye conjunctiva." International Journal of Imaging Systems and Technology 30.1 (2020): 112–125.
26. Magdalena, Rita, et al.: "Convolutional neural network for anemia detection based on conjunctiva palpebral images." Jurnal Teknik Informatika (Jutif) 3.2 (2022): 349–354.
27. Agrawal, Akhil.: "Detecting Anemia from retinal images using Deep Learning." (2021).

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.